

Four problems, and the scaffolding falls away

One gene, made at rate k and removed at rate γ per molecule. The count n is a whole number that changes by one at a time.

Today's three results, and nothing else.

$$\text{birth-death: } \langle n \rangle = \sigma^2 = \frac{k}{\gamma}, \quad \text{Fano} = \frac{\sigma^2}{\langle n \rangle} = 1, \quad \eta^2 = \frac{\sigma^2}{\langle n \rangle^2} = \frac{1}{\langle n \rangle}$$

$$\text{bursts of } b \text{ proteins per mRNA: } \text{Fano} \approx 1 + b$$

$$\text{self-repression at the same mean: } \text{Fano} = \frac{1}{1+g}, \quad g = \frac{n u^n}{1+u^n}, \quad u = \frac{x}{K}$$

Two identities do all the bookkeeping: $\eta^2 = \text{Fano} / \langle n \rangle$, and noise sources that are independent add as squares, $\eta_{\text{tot}}^2 = \eta_{\text{int}}^2 + \eta_{\text{ext}}^2$. At each transition this sheet asks *why does that step follow?* Answer in writing before you go on.

1 — Fully worked: an unregulated gene at a mean of 25

Take $k = 25$ per lifetime and $\gamma = 1$ per lifetime. The stationary distribution is Poisson, so

$$\langle n \rangle = \sigma^2 = 25, \quad \text{Fano} = \frac{\sigma^2}{\langle n \rangle} = 1,$$

$$\eta^2 = \frac{\text{Fano}}{\langle n \rangle} = \frac{1}{25}, \quad \eta = 0.20.$$

Going through the Fano factor looks like a detour here, because it is 1. It is the route every other item takes, so it is worth walking once while the answer is known.

Now double the promoter, $k = 50$. The mean doubles, $\eta^2 = 1/50$, and $\eta = 0.141$: the noise falls by a factor of 1.41, not 2. *Why does that step follow?* — because the spread of a count grows as the square root of the count, so the *relative* spread shrinks only as one over its square root.

2 — One step given: the same mean, made in bursts

Keep the mean at 25 proteins, but make them from a weak promoter and a strong RBS: each mRNA now makes $b = 4$ proteins on average before it decays, and the mRNA is short-lived compared with the protein.

$$\text{Fano} \approx 1 + b = \underline{\hspace{2cm}}, \quad \eta^2 = \frac{\text{Fano}}{\langle n \rangle} = \underline{\hspace{2cm}}, \quad \eta = \underline{\hspace{2cm}}$$

Why does that step follow? — same mean, same promoter-times-RBS product. Say in one sentence what changed that the mean cannot see.

Which would you build if you needed this protein at 25 per cell with as little cell-to-cell variation as possible, and what would you change to get there?

3 — Nothing given: the same gene, repressing itself

Back to one protein at a time (no bursts), now at a mean of 100. Make the gene repress itself with cooperativity $n = 3$, and tune the promoter so the steady state sits at $x = K$, i.e. $u = 1$ — the same mean as the constitutive gene you are about to compare it with.

$$g = \frac{n u^n}{1 + u^n} = \text{____}, \quad \text{Fano} = \frac{1}{1 + g} = \text{____}$$

Compare η with the constitutive gene at the same mean of 100:

$$\eta_{\text{constitutive}} = \text{____}, \quad \eta_{\text{self-repressed}} = \text{____}$$

Why does that step follow? — the feedback did not change how often molecules are made or destroyed. What did it change?

4 — Bare problem: read Table 1

Elowitz et al. 2002, Table 1 (p. 1185), noise in units of 10^{-2} :

STRAIN	INTENSITY	η_{int}	η_{ext}	η_{tot}
M22, constitutive	1	5.5	5.4	7.7
RP22, LacI-repressed	0.030	25	33	41
RP22 + IPTG	1.00	6.3	9.8	11.7
M22 + repressilator	0.18	12	42	43

Pick **two rows** whose comparison tells you something. Say which column moved, by how much, and what in the cell moved it. Then check one row against $\eta_{\text{tot}}^2 = \eta_{\text{int}}^2 + \eta_{\text{ext}}^2$.
